

XIAOYANG WAN

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Education

University of Pennsylvania

M.S. in Computer and Information Science

Sep. 2025 – May 2027

Philadelphia, PA

Zhejiang University

B.S. in Computer Science and Technology; GPA: 3.96/4.00

Sep. 2021 – May 2025

Zhejiang, China

Professional Experience

Upstart

Machine Learning Engineer Intern

Jun. 2026 – Aug. 2026

San Jose, CA

- Optimized a **Metaflow** transaction-embedding pipeline via **16 single-GPU BERT** shards and indexed **Parquet** reads; validation: **2x** speedup, **80%** lower cost, **90%** less peak shard RAM.
- Replaced **NumPy** scans with hash lookups over 7M trade accounts; cut whitelist filtering **875s to 11s**.
- Traced bin flips in **2.13%** of borrower records to 1-ULP **NumPy log1p** drift across CPU and library versions; froze raw-amount thresholds for **1e-5** embedding parity across hosts.
- Fixed **AWS Batch** GPU contention via device pinning and **16-min** TCP stalls via a shared **@batch** timeout.
- Benchmarked six-bag **PyTorch** GPU training with cached feature matrices and parallel prep; cut runtime from **3h21m to 1h38m**, compute cost **73%**, and prep time **37%**.

Molardata & 2077 AI

Machine Learning Engineer Intern

Apr. 2025 – Aug. 2025

Zhejiang, China

- Built a **LangChain** generator-critic-judge pipeline for adversarial tests on **2,700** Codeforces problems: **GPT-4o** wrote generator scripts, **GPT-4o-mini** flagged missed attack types, and a GPT-4o judge scored coverage.
- Validated tests with an input verifier plus reference and known-wrong solutions (near-**100%** error detection in a **20-problem** pilot); ran up to three feedback rounds with per-step **checkpoints** for resumable runs.
- Auto-validated **1,700** problems; **200/400** reviewed test suites passed subsequent human quality checks.
- Built a **FLUX** three-turn editing MVP for **200** image sequences: GPT-4 wrote edit prompts and target captions, a vision model captioned each result, and an **LLM judge** checked it against the target caption.
- Achieved **83%** manual-review pass rate on 200 sequences; refined prompts to catch left-right position swaps.

DCD Lab, Zhejiang University

Research Assistant

Jun. 2024 – Jan. 2025

Zhejiang, China

- Adapted and ran seven existing image-editing data pipelines (add, remove, move, background, material, tone, rotation), producing **500K+** of the **2.5M** AnyEdit image pairs.
- Split editing instructions into index ranges across **two GPUs** and used skip-if-exists **checkpointing**, so interrupted batches resumed without regenerating finished images.
- Evaluated generated edits with **CLIP**, **DINO**, and L1 metrics.

Selected Project

Multimodal RAG: Anime-to-Manga Retrieval | *Student Research MVP with Hongjun Liu*

Jan. 2024 – Apr. 2024

- Implemented an anime-to-manga **RAG** MVP: embedded manga pages with **CLIP ViT-L/14** and stored the vectors in Docker-hosted **Milvus** with page descriptions and metadata for cosine search.
- Crawled **100+** annotated manga pages spanning **75 chapters** from a manga wiki; dropped non-story pages.
- Fed the **top-5** pages and their descriptions to **GPT-4V** for plot descriptions; built a **Gradio** demo.

Publication

CVPR 2025 Oral | *AnyEdit: Mastering Unified High-Quality Image Editing for Any Idea* | Co-author.

Technical Skills

Programming & Systems: Python, C++, Go, SQL, Linux, Git, Multithreading, TCP Networking

Machine Learning: PyTorch, scikit-learn, Transformers, BERT, MPNet, FastText, GPU Inference, Model Training

Data Engineering: NumPy, pandas, Parquet/PyArrow, Feature Engineering, Batch Processing, Vectorization, Caching

LLM & Vision: LangChain, Multi-Agent Systems, Prompt Engineering, LLM Evaluation, CLIP, Diffusion Models, Milvus

Cloud & MLOps: AWS Batch, EC2, S3, Metaflow, MLflow, Docker, Workflow Monitoring, Checkpointing